Obsidian Second Brain

v29 · 2026-10-12

1 · INPUTS

Obsidian Web Clipper
Browser + iOS/iPadOS share sheet

Any website articles including paywalled sources.
→ Clippings/ (Clipping Template)

Voice Note (iPhone)
iOS Shortcut "Obsidian Note"

Dictate → signed email → intake mailbox
→ Creations/ (via Python voice-cleanup)
mail pulled ≤5 min, then 10-second watcher

Markitdown Dropper
PySide6 drop-zone app + cleanup pass

.docx · .pptx · .xlsx · .pdf · audio · image
→ Creations/ · images → Z_attachments/
extracts images · normalizes bullets · adds YAML

People Create
QuickAdd

Title · Org · Email · Mobile · Photo
→ People/ (links Categories/People)

Meeting Notes
Meeting / Meeting Type templates

Live note-taking during meetings
→ Meetings/ (then → Meetings/History/)

Hand-written / Creations
Note Template · Journal

→ Creations/ · topical folders

YouTube Summarizer
Share Sheet → Claude (Sonnet 5.5)

→ Clippings/YouTube/ (yt-dlp transcript → markdown)

Podcast Transcribe
Episode · RSS feed · Apple Podcasts URL

Share → signed email, or drop an audio file
→ Clippings/ (verbatim transcript, then tagged)
local Whisper · 900-sec tick · 1 episode per run

2 · AUTOMATION

Templates · QuickAdd
Clipping · Voice · People · Meeting
Note · Journal · Move-to-Knowledge
applied at create time, in-app

LaunchAgent · Voice Cleanup
com.voice-cleanup.plist
voice_cleanup.py · 10-sec poll
Watches ~/SourceMedia/VoiceInput/
Calls Claude API → polish + YAML
Writes to Creations/, deletes raw
~1 API call per voice note

LaunchAgent · Strip Ads
com.obsidian.strip-ads
strip_ads.py · WatchPaths trigger
Watches Clippings/ (~5-sec throttle)
Strips NYT/WSJ ad blocks & iframes
no API call · purely local regex

Python LaunchAgent
com.tag-clippings
Runs at :00 and :30 (every 30 min)
Scans Clippings, Creations, Meetings
Allowlist tags · .tag_tracking.json
(body-only · Tag Taxonomy.md)
claude-haiku-4.5 · 1 API call/note

Manual curation
After tag review · Move to Knowledge
Meetings/ → Meetings/History/

3 · VAULT STORAGE (~/Obsidian)

~/Obsidian (single consolidated vault)

CAPTURE (scanned by tagger)

Clippings/	web articles
Creations/	voice + writing + markitdown
Meetings/	active meeting notes
↳ Meetings/History/	archived (still tagged)
People/	person notes

REFERENCE & STRUCTURE

Knowledge/	curated notes
Categories/	maps of content
Topics/	subject hubs
Groups/	people groupings
Actions/	to-dos & lists
Daily/	daily notes
Syntheses/	position notes (third tier)

SYSTEM

Templates/	+ Templates/Scripts/ (Python)
Z_attachments/	binary attachments
Z_archive/	completed to-dos & deprecated materials
.tag_tracking.json	processed-file ledger

SYNC & REPLICATION · end-to-end encrypted

Obsidian Sync	E2E encrypted in transit & at rest
Devices: Mac · Windows · iPhone · iPad	read/write everywhere
Vault encryption key in OS keychain only	never leaves device

API keys in the OS keystore, outside the vault — not synced

DATA CLASSIFICATION · policy-aligned

public · internal-use-only · confidential · restricted

Required `classification` frontmatter on every note · set at create-time by Templates & ingest pipelines (add_classification.py backfill)
restricted tier is excluded from the local RAG sync

4 · ORGANIZATION LAYER

Categories (MOCs)
Maps of content (MOCs)

e.g., People · Projects · Companies · Places · Meetings · Clippings · Journal · …
One page per content type, with a Bases index

Topics
Subject hubs that aggregate notes

One page per subject area, Bases-driven

Groups
People groupings & rosters

One page per recurring group, with member links

Tags (hierarchical)
Semantic tag taxonomy

Applied every 30 min by the com.tag-clippings LaunchAgent

Knowledge graph
Backlinks + Bases views

Person ↔ Meetings ↔ Topics ↔ Groups
Dashboards surface live views

Retrieval
Omnisearch · Tag Wrangler · Bases
Manual review & "Move to Knowledge"
Curated promotion → Knowledge/
Manual archive → Meetings/History/

Syntheses (third tier)
Position notes anchored to Knowledge

Hand-authored or LLM-assisted; one note per defensible position with backlinks to sources

Prompts: Convergence · Divergence · Gap · Brief

→ Syntheses/ · surfaced via Topics/Syntheses

5 · SOURCE MAIL TRANSPORT · phone → Mac

replaces iCloud Drive — no cloud-drive client on any machine, Mac or Windows

authenticated payloads · HMAC-SHA256

Producer · iPhone / iPad / Mac
Shortcut → Scriptable signs (Web Crypto)
One plain-text email per capture
Types: voice · podcast
Token in Scriptable Keychain — per device, never syncs; run the signer standalone once
macOS: Run Shell Script → --emit (no helper)
one From address across devices (allowlist)

Relay · dedicated intake mailbox
Unguessable address, not a +alias
2-Step Verification + App Password
One App Password per machine (revocable)
Body: v · type · ts · payload · hmac
MAC covers a length-prefixed canonical string
First occurrence of a key wins
Subject is cosmetic — never used for routing

LaunchAgent · Source Mail Pull
com.obsidian.source-mail-pull
source_mail_pull.py · IMAP · every 300 s
HMAC-SHA256 — the token is never transmitted
Sender allowlist, fails closed · 7-day replay bound
Route type read from inside the MAC
Filenames generated locally · text/plain, size-capped
rejected drops are marked Seen — just re-share

~/SourceMedia · local drop root
VoiceInput/ → voice_cleanup.py
PodcastInput/ → podcast_watch.py
Subfolders, not one flat queue — all drops are .txt, so the watchers would otherwise race
Outside the vault · not in Obsidian Sync scope
drop-folder contract unchanged — watchers untouched

6 · SCHEDULED JOBS & DERIVED SURFACES

launchd timers that operate on the vault — none ingest new content

canonical: ~/Obsidian/Templates/Scripts

Morning Dashboard
com.morning-dashboard
7:00 AM · Mon–Fri · launchd
Open to-dos · today's meetings · new clippings + creations since midnight
→ ~/Obsidian/Z_dashboards/ (HTML, opens in Chrome)
local-only · no API call

Meeting Prep · follow-ups
com.obsidian.meeting-prep
every 5 min, weekday business hours
Grafts open [ ] tasks per attendee into today's Individual meeting notes
Idempotent HTML-comment fence
vault-scope · no network

Meeting Pre-Population
Producer: meeting_pull.py (headless Claude)
M365 connector · weekdays 05:00 → ~/MeetingIngest
Consumer: meeting_prepopulate.py
com.meeting-prepopulate · Sun 21:00 + WatchPaths + poll
Builds upcoming meeting notes
idempotent via source_uid frontmatter

Group Photo Refresh
nightly · 02:00 (two-script pass)
1 · insert_group_placeholders.py
prefixes bare member links with a placeholder
2 · refresh_groups.py
swaps placeholder → real headshot when it exists
stdlib-only · no API call

7 · OPTIONAL — Local LLM with Vault as RAG

Requires capable hardware (Apple Silicon Mac or equivalent). Not part of core workflow.

on-device · no cloud calls

Inference & UI
Ollama (MLX runtime)
MLX auto-selected on Apple Silicon; runs existing GGUFs.
Open WebUI (Docker)
Bound to 127.0.0.1:3000 · signups locked
Admin Panel · Knowledge · Custom Models
RAG node: primary Apple Silicon Mac only

Custom Models (each + Vault)
Llama 8B + Vault — quick lookups (~8 GB)
Gemma 4 26B + Vault — daily default · MoE (~16 GB)
Llama 70B + Vault — synthesis (~50 GB)
Nemotron + Vault — grounding-tuned (~86 GB)
Each "+ Vault" pre-attaches the Knowledge collection

RAG over the vault
Embedder: nomic-embed-text (Ollama-pulled)
Retrieval: hybrid search · Top K = 6
Chunking: Markdown header splitter
Reranker: bge-reranker-v2-m3 (optional)
Sync: Templates/Scripts/obsidian-rag-sync.py
Nightly LaunchAgent 03:15 · pushes to Open WebUI Knowledge
Indexes ~/Obsidian; profile note anchors role context

Access & security
Localhost only (127.0.0.1:3000)
No supported remote-access path
Hardened posture
Ollama firewall-blocked (no LAN exposure)
Open WebUI bound to localhost only
Secondary node: personal models via LM Studio + LM Link (no vault)

8 · SECURITY CONTROLS & GATES

tripwires, not boundaries: they run as the user and report what changes

alerts → notification + alerts.log

Plugin Integrity Check
com.obsidian.security.plugin-check
on change to plugins · HMAC-signed allowlist
Pinned releases, per-file SHA-256
Code-running settings watched
Re-vet by the operator after any update
local · no network

Workflow Integrity Monitor
com.obsidian.security.integrity
on change + daily · hash baseline
Scripts · agents · venv · templates
job settings · Claude config · secrets file
Baseline adopted only on known drift
fails closed on an unreadable scan

Classification & Export Gate
com.obsidian.classify · daily
L0 detectors · L1 model · L2 human review
Proposes tiers; never lowers one
Export gate fails closed on what it cannot read or classify
restricted never leaves the vault

Ingest Guard & Standing Checks
Outside text cannot trigger plugin code
applied by every pipeline that writes
Seven security passes on every push
Public-tree gates: classification + identity
Installed git hooks must be current
artifacts outside the repo

Architecture rationale

Each input lands in its target capture folder under ~/Obsidian (destinations noted inside each input box). Ingest LaunchAgents do mechanical, deterministic cleanup at the edge: voice_cleanup polishes dictations (~1 Claude API call each); strip_ads removes NYT/WSJ ad blocks (no API); YouTube share-sheet captures route through the Claude summarizer; podcast_watch transcribes drops locally (MLX Whisper → faster-whisper → ONNX, one episode per 900-sec tick) into Clippings/; com.tag-clippings auto-tags every 30 min. Articles are clipped by hand with the Obsidian Web Clipper in a signed-in browser — no automation, no stored credentials.

iCloud Drive is no longer a dependency anywhere in the workflow. Phone-originated captures (band 5) now reach the Mac by mail: the device signs a plain-text payload with HMAC-SHA256 and emails it to a dedicated intake mailbox; com.obsidian.source-mail-pull fetches it over IMAP every 300 s and writes it into ~/SourceMedia/{VoiceInput,PodcastInput}/. This separates the drop-folder contract from the transport, so the watchers were unchanged — and it removes Apple iCloud for Windows from machines that otherwise need nothing from Apple. A mailbox is world-writable, which is why the payload carries its own proof: unguessable address, sender allowlist, MAC over a length-prefixed canonical string, routing type taken from inside the MAC, a 7-day replay bound, and locally generated filenames. The token itself is never transmitted.

A second class of jobs (band 6) operates ON the vault rather than ingesting: com.morning-dashboard builds a 7 AM weekday HTML surface; meeting-prep grafts attendee follow-ups into today's notes; meeting-prepopulate consumes a schedule handoff from meeting_pull.py, a headless Claude session on the Microsoft 365 connector, via ~/MeetingIngest outside the vault; group-photo refresh runs nightly. Band 8 watches all of it.
Every note now carries a policy-aligned `classification` property (public / internal-use-only / confidential / restricted), set at create-time; restricted is excluded from the local RAG sync. All automation scripts are canonically located at ~/Obsidian/Templates/Scripts. Vault content is replicated cross-device via Obsidian Sync with E2E encryption (key in the local OS keychain only).
API keys stay in the OS keystore; code repos and ~/SourceMedia sit outside the sync scope. Band 8's controls are tripwires: they report a change, they do not prevent it.